UX DESIGN - LECTURE 01

1. Purpose of UX Design
UX (User Experience) design focuses on creating digital products that are **easy, efficient, and pleasant for users to interact with**.
Key goals:
- Help users achieve their goals **quickly and easily**
- Make products **usable, intuitive, and valuable**
- Improve **customer satisfaction and engagement**
- Provide **competitive advantage** for digital products

If a product is difficult to use, users will likely **abandon it regardless of its features or technical quality**.

2. Digital Product
A **digital product** is any software that runs on a digital device.
Examples:
- Mobile apps
- Websites
- Computer software
- Video games
- Web browsers
- Smartwatch apps

Devices include:
- Computers
- Smartphones
- Tablets
- Smart devices

3. Usability
Usability measures how easily users can use a product to achieve their goals.
Core aspects:
- Ease of navigation
- Accessibility of information
- Speed of task completion
- Simplicity of interaction

Good usability means:
- Users can **quickly find what they need**
- The product **reduces effort**
- The product **improves the user's life or workflow**

Poor usability leads to frustration and abandonment.

4. User Experience (UX)
User Experience (UX) refers to the **overall interaction between a user and a digital product**.
UX considers:
- Ease of use
- Efficiency
- Value gained from the product
- Speed of completing tasks
- Emotional satisfaction

Effective UX ensures:
- Users reach their goals **with minimal effort**
- The interaction is **smooth and intuitive**

5. Example of Poor Usability
Scenario: Emergency services app.
Problem:
- User opens the app during an emergency
- Instead of contacting help immediately, the app asks the user to fill out a survey

Issue:
- The design **blocks the user's main goal**

Lesson: UX design must **prioritize the user's primary objective**.

6. UX Design

UX Design is the process of designing digital products to maximize usability and user satisfaction.

Design elements involved:

- Buttons
- Input fields
- Navigation menus
- Links
- Scroll bars
- Content layout

It involves:

- Structuring interfaces logically
- Reducing user effort
- Creating intuitive interactions
- Delivering clear value

Goal: Make the interaction **fast, intuitive, and pleasant**.

7. UX Design Does Not Require Artistic Skill

UX includes multiple disciplines beyond visual design.

- Examples: UX research, UX strategy, UX evaluation

Many UX roles focus on **understanding users rather than designing visuals**.

8. UX Research

UX research identifies **who the users are and what they need**.

Research focuses on:

- Target user groups
- User needs
- Pain points
- Problems users want solved
- Desired product features

Methods:

- User interviews
- Observations
- surveys
- competitor analysis

Purpose: Design products **based on real user needs rather than assumptions**.

Skipping research often leads to **product failure**.

9. Prototyping

A **prototype** is an early version of a product used for testing ideas.

Reasons for prototyping:

- Faster testing
- Lower development cost
- Early feedback from users

Types:

Low-fidelity prototypes

- Rough sketches
- Paper drawings
- Simple layouts

Advantages:

- Easy to change
- Encourages honest feedback

High-fidelity prototypes

- More detailed digital versions
- Similar to the final product
- Built using tools like **Figma or Adobe XD**

Purpose: Validate ideas **before full development begins**.

10. UX Strategy

UX strategy defines the **long-term vision for the user experience**.

It connects:

- User value, Business value, Product goals

Key aspects:

- Define the current user experience
- Define the desired experience
- Plan how to transition between them

UX strategy ensures:

- The product benefits **both users and the business**
- The experience is consistent across all touchpoints.

11. User Interface (UI) Design

UI design focuses on the visual presentation of the product.

- UI includes: Colors, Typography, Images, Shapes, Layout, Interface structure
- UI design ensures: Clear visual hierarchy, Attractive design, Efficient interaction

UX defines **how the product works**,
UI defines **how the product looks**.

12. Product Management

Product management oversees the **entire lifecycle of a digital product**.
Responsibilities:

- Product planning
- Design coordination
- Launch
- Maintenance
- Continuous improvement

UX knowledge is important for:

- Prioritizing features
- Improving user satisfaction
- Managing product development.

Product managers often work closely with UX designers.

13. UX Design Process

UX follows a structured but flexible process.

1. Understand

Develop empathy for users.
Focus: User emotions, motivations, psychology, behavioral patterns

2. Research

Identify user needs and problems.
Activities:

- User analysis
- competitor evaluation
- identifying motivations and contexts

3. Idea Generation

Create solutions to user problems.
Methods:

- brainstorming, collaboration
- co-creation with users or stakeholders

Goal: Generate multiple ideas before selecting the best solution.

4. Prototyping

Create a mock version of the product.
Start with:

- simple sketches
- wireframes

Then move to:

- detailed interactive prototypes.

5. Testing

Test prototypes with users.
Testing reveals:

- usability issues
- confusing features
- unexpected behaviors

If problems appear:

- return to research or ideation.

6. Release

Developers build and launch the final product.
Activities:

- coding
- integration
- system compatibility testing

7. Continuous Analysis
UX work continues after launch.
- Actions: monitor usage, gather feedback, release improvements, create updates

Products are **never final**.

14. Iterative Nature of UX
The UX process is **not linear**.
Designers often move back and forth between stages.
Example loop: Research → Idea → Prototype → Test → Refine → Test again.
This iterative approach improves product quality.

15. Collaboration Tools
Modern UX work supports remote collaboration.
Capabilities:
- Real-time collaboration
- link sharing
- remote feedback
- prototype testing.

Example tools:
- **Figma**
- Adobe XD

16. Role of Developers
UX designers focus on **experience design**, not coding.
Developers:
- implement the product
- build technical systems
- perform compatibility testing

However, UX designers may learn coding or use **no-code tools**.
- Examples: Wix, Shopify, Microsoft Power Apps

17. Importance of Human Psychology in UX
Understanding users requires knowledge of:
- human psychology
- personality traits
- decision-making behavior
- motivation patterns

This helps designers predict **how users interact with products**.

18. Applications of UX Knowledge
UX knowledge is valuable across many fields.
Relevant professions:
- software developers
- marketers
- product managers
- graphic designers
- entrepreneurs
- corporate leaders
- politicians

UX helps improve:
- customer journeys
- digital marketing funnels
- operational efficiency.

19. UX in Marketing
UX explains user behavior during digital marketing processes.

Example insights:

- why users abandon websites, why sales funnels fail, how to improve conversions

UX helps marketers design **better user journeys**.

20. UX in Technology Development

Technology becomes valuable only when users can benefit from it.
UX helps:

- transform technology into usable products
- commercialize innovation
- connect technical capabilities with user needs.

21. UX as a Career

UX design is a **growing and versatile profession**.

- Career characteristics: remote work opportunities, freelance possibilities, flexible working structures, high global demand
- Entry requirements: UX skills, strong portfolio

Formal degrees are **not mandatory**.

22. Future of UX

UX demand is increasing due to emerging technologies.
Key areas:

- Internet of Things (IoT), Virtual Reality (VR), Augmented Reality (AR), Voice interfaces, Artificial Intelligence systems

All require **human-friendly interfaces**.

23. UX Design Principles Demonstrated

Effective UX requires:
Clear layout and hierarchy: Information should be visually structured.
Concise communication: Messages should be short and direct.
Logical flow: Interface elements should appear in a natural order.
These improve clarity and usability.

24. The Core Takeaways

UX design focuses on **humanizing technology**.
Purpose:

- make digital systems understandable
- reduce user effort
- improve people's daily interactions with technology.

UX DESIGN - LECTURE 02

1. UX Specialist Review
A **UX specialist review** is the process of evaluating a digital product (app, website, or prototype) to identify **usability problems and improvement opportunities**.
Other names for the same concept:
- UX evaluation, UX audit, Cognitive walkthrough ,Heuristic evaluation

Purpose:
- Inspect a design using **UX principles and best practices**
- Identify usability issues **before user testing**
- Provide **recommendations for improvement**

Characteristics:
- Fast and cost-effective
- Based on expert judgment
- Does **not replace user testing**, but complements it.

2. Who Can Conduct a UX Specialist Review
Anyone with experience using digital products can participate, but effectiveness varies.
Levels of evaluators:
General digital users
- Familiar with apps and websites
- Can participate in **user testing**
- Limited understanding of UX theory.

UX practitioners
More effective reviewers because they have:
- UX design knowledge
- usability principles
- research experience
- pattern recognition from previous projects.

Helpful backgrounds:
- UX design
- UX research
- psychology
- consumer research.

3. Effectiveness of Multiple Evaluators
Research by **Nielsen and Landauer** shows that different numbers of evaluators detect different percentages of usability issues.

Evaluators	1	2	3	4	5	10	15
Issues Found	~33%	~50%	~60%	~70%	~74%	~86%	~92%

Key insight:
- Each additional evaluator finds **fewer new issues**.

This is known as **diminishing returns**.

4. Optimal Number of Evaluators
Research suggests **four evaluators** is often the best balance between cost and benefit.
Reason:
- High percentage of issues detected
- Costs remain manageable.

Practical alternative if resources are limited:

1. Hire **one experienced UX reviewer**
2. Have **three internal team members review the report**
3. Discuss findings collaboratively.

This method:

- improves evaluation quality
- transfers UX knowledge to the team.

5. Importance of Mobile UX

Mobile usage dominates digital interaction.

Observation:

- People check their phones **every 5–6 minutes** on average.

Implication for UX:

- Mobile interfaces must be **carefully designed**
- Responsive design is critical
- Mobile experience must be prioritized.

6. Steps to Conduct a UX Specialist Review

Step 1 — Identify Business Objectives

Understand the purpose of the product.

Methods:

- interviews with stakeholders
- surveys with company leaders

Important practice:

- confirm findings with the client.

Goals:

- identify strengths of the product
- identify weaknesses
- understand business priorities.

Step 2 — Understand the Users

Identify who the users are and how they behave.

- Approaches: User research & User personas (fictional profiles representing user types)
- Activities: Observe user behavior, Collect feedback & Study expectations and experiences.
- Consistency rule: All participants should follow the **same testing procedure**.

Step 3 — Collect UX Data

Use analytics to understand user behavior.

- Tools mentioned: Google Analytics, Kissmetrics, Crazy Egg, GoodData, Hotjar, Mixpanel
- Data insights include: who uses the product, why they use it, how they interact with it.

Step 4 — Identify UX Issues

Analyze the product for usability problems.

Common areas to evaluate:

- **Responsiveness:** Interface should adapt to different devices and screen sizes.
- **Message clarity:** Text should be clear and understandable.
- **Typographic hierarchy:** Information should appear in priority order: Heading, Subheading, Body text.
    - Common issues: long text blocks, confusing instructions, buttons distracting from key elements.
- A**ccessibility:** The product should be usable by all users, including those with disabilities.
    - Example tool: Google Lighthouse.

Step 5 — Propose UX Improvements

After identifying issues, propose solutions.

- UX designers act as: Problem solvers, Experience optimizers.

Solutions may involve:

- Interface redesign, Workflow changes, Clearer messaging, Improved accessibility.

Step 6 — Compile the UX Report

Prepare a formal UX evaluation report.

Report contents:

- findings
- identified usability issues
- proposed solutions
- improvement opportunities.

Characteristics of a good report:

- clear structure
- actionable recommendations
- aligned with business goals.

Templates can help but should be **adapted for each project**.

7. UX Evaluation Frameworks

UX reviews often use **structured frameworks** or checklists.

Examples:

- Nielsen's usability heuristics
- Laws of UX (John Jablonski)

These frameworks help designers systematically analyze interfaces.

8. Nielsen's 10 Usability Heuristics

A widely used set of principles for designing usable interfaces.

They act as **rules of thumb for evaluating UX**.

9. Heuristic 1 — Visibility of System Status

Users should always know:

- what is happening
- where they are in a process.

Without feedback:

- users may repeat actions unnecessarily.

Examples:

- loading indicators
- progress bars
- button feedback after clicking.

10. Heuristic 2 — Match Between System and Real World

Interfaces should use **familiar concepts and visual cues**.

Examples:

- Buttons that look clickable
- Warning colors like red or orange
- Icons resembling real objects.

Goal:

- Reduce cognitive effort.

11. Heuristic 3 — User Control and Freedom

Users should feel in control.

Important features:

- back buttons
- undo / redo
- cancel options
- previews before committing actions.

Purpose:

- allow exploration without fear of mistakes.

12. Heuristic 4 — Consistency and Standards
Design elements should follow **common patterns users already understand**.
Examples:
- blue underlined text for links
- back button on the left
- red indicating danger.

Consistency reduces learning effort.

13. Heuristic 5 — Error Prevention
Design should prevent mistakes before they occur.
Examples:
- warning when transferring money to the wrong account
- notification when navigating to a closed location.

Goal: guide users toward correct actions.

14. Heuristic 6 — Recognition Rather Than Recall
Users should not rely on memory.
Design should present options clearly.
Examples:
- dropdown lists
- visible icons
- save buttons instead of requiring keyboard shortcuts.

Recognition is easier than recall.

15. Heuristic 7 — Flexibility and Efficiency of Use
Interfaces should support both:

beginners
- simple guidance
- clear navigation.

experienced users
- shortcuts
- personalization
- faster workflows.

16. Heuristic 8 — Aesthetic and Minimalist Design
Interfaces should be:
- visually appealing
- clean
- uncluttered.

Minimalism improves usability.

Problems with clutter:
- distracts users
- causes cognitive overload
- hides important information.

17. Heuristic 9 — Help Users Recover From Errors
Since users will make mistakes, the system should help them recover.
Requirements:
- clear error messages
- understandable explanations
- suggestions for fixing the problem.

Example:
- spelling suggestions
- instructions for opening unsupported file types.

18. Heuristic 10 — Help and Documentation
Users should always have access to assistance.
Forms of help:
- live chat, support phone numbers, tooltips, FAQs, guides, contextual help messages.

Goal: Provide help **when users get stuck**.

19. Icon Design Principle
Icons should represent real-world concepts so users can recognize them easily.
Testing method:

- create icons representing concepts
- show them to users
- check whether they understand the meaning.

This ensures **real-world recognition**.

20. The Core Takeaways
Important points to remember:

- UX specialist reviews identify usability issues quickly and cheaply.
- They rely on **UX principles and expert evaluation**.
- They complement but **do not replace user testing**.
- Around **4 evaluators** provide an optimal balance between cost and effectiveness.
- UX evaluation should consider **business goals, user needs, analytics, and usability issues**.
- Findings must be documented in a **structured UX report**.
- Nielsen's **10 usability heuristics** provide a practical framework for evaluating interfaces.
- Mobile experience and responsive design are critical in modern UX.
- UX designers function primarily as **problem solvers and experience optimizers**.

UX DESIGN - LECTURE 03

1. Purpose of Nielsen's UI Principles
Jacob Nielsen's **10 usability heuristics** are guidelines used to evaluate whether a digital interface is **intuitive, efficient, and user-friendly**.
Purpose:
- Provide a **framework for evaluating interfaces**
- Identify usability issues early
- Improve interaction between users and digital systems

These principles function as **rules of thumb**, not strict laws.

2. The 10 Core UI Principles
1. Recognition rather than recall
2. Flexibility and efficiency of use
3. Aesthetic and minimalist design
4. Match between system and real world
5. Help users recognize, diagnose, and recover from errors
6. Help and documentation
7. User control and freedom
8. Visibility of system status
9. Consistency and standards
10. Error prevention

3. Visibility of System Status
The system must always **inform users about what is happening**.
Key ideas:
- Immediate feedback after user actions
- Continuous communication between system and user

Examples:
- button animation after click
- loading spinners
- progress bars
- step indicators
- status messages
- highlighted active pages

Time feedback guidelines:
- Short tasks → spinner/loading animation
- Medium tasks (~9+ seconds) → progress bar
- Long tasks (~1 minute+) → time estimation

Benefits:
- builds trust, prevents repeated actions
- allows users to predict system behavior.

4. Match Between System and the Real World
Interfaces should use **concepts familiar to users from the real world**.
- Principles: Use natural language, avoid internal or technical jargon, Follow logical information flow.

Two important concepts:
Natural Mapping: Design elements reflect real-world spatial logic.
- Examples: volume up = upward slider, back button = left arrow, next button = right arrow.

Skeuomorphism: Digital elements visually resemble real objects.
- Examples: Trash can icon for delete, Credit card icon for payment, Pencil icon for edit.

Important consideration: Users' interpretation of symbols varies based on:
- culture, language, age, profession

UX research helps determine the correct metaphors.

5. User Control and Freedom
Users must feel **free to explore without fear of irreversible mistakes**.

Essential features:

- back buttons, cancel actions, undo / redo, navigation controls, exit options.

Advanced control mechanisms:

- version history, ability to revert to earlier versions, cancel ongoing processes.

Benefits:

- reduces anxiety, improves confidence, encourages exploration.

6. Consistency and Standards

Interfaces should behave in **predictable and familiar ways**.

Two types of consistency:

Internal consistency

Consistency within the same product.

Examples:

- same navigation structure across pages
- similar gestures across screens.

External consistency

Consistency with industry standards.

Examples:

- gear icon for settings
- magnifying glass for search
- blue underlined text for links.

Importance:

- reduces learning effort
- increases intuitive usability.

7. Error Prevention

Design systems to **prevent mistakes before they occur**.

Two error types:

Slips

Accidental mistakes caused by lack of attention.

Mistakes

Errors caused by misunderstanding the system.

Error prevention strategies:

- limit invalid inputs
- use constraints
- provide smart defaults
- confirm risky actions
- reduce cognitive overload.

Examples:

- preventing checkout date earlier than check-in
- Gmail warning about missing attachments
- numeric input validation.

8. Recognition Rather Than Recall

Users should **recognize options instead of remembering them**.

Why:

- recognition requires less mental effort than memory recall.

Examples:

- dropdown menus
- icons with labels
- visible navigation menus
- saved passwords
- autofill forms.

Practical implementations:

- combine icons with descriptive text
- provide search functionality
- provide filters for large data sets.

Goal: Minimize reliance on memory.

9. Flexibility and Efficiency of Use
Interfaces should work well for **both beginners and advanced users**.
Approaches:

Personalization
Users customize interface layout or settings.
Examples:
- rearranging dashboard widgets
- customizing toolbars.

Accelerators
Features that increase speed for experienced users.
Examples:
- keyboard shortcuts, gestures (swipe, shake)
- bookmarks, tagging.

Multiple interaction paths
Users can accomplish the same task through different workflows.
Example: Different ways to make a payment in a banking app.
Benefits:
- improves productivity
- accommodates different user preferences.

10. Aesthetic and Minimalist Design
Interfaces should be **clean, simple, and focused**.

Problems caused by clutter:
- slower navigation
- cognitive overload
- user frustration
- task abandonment.

Principles:
- remove unnecessary elements
- reduce visual noise
- highlight essential content.

Design rule: Ask **what can be removed**, not what can be added.
Techniques:
- grouping information
- clear headings
- logical content hierarchy
- visual scanning structure.

11. Helping Users Recognize and Recover from Errors
When errors occur, systems must help users **identify, understand, and fix them**.

Effective error handling process:
1. Inform user an error occurred
2. Explain what the problem is
3. Provide a solution
4. Track or follow up on resolution
5. Improve the system to prevent future occurrences.

Good error messages should:
- be clear
- use plain language
- avoid technical codes
- suggest corrective action.

Example implementations:
- spelling suggestions
- password requirement checklists
- automated fixes.

12. Help and Documentation
Even intuitive systems may require assistance for complex tasks.
Forms of help: tooltips, FAQs, tutorials, contextual hints, help articles, live chat, phone support.
Good documentation characteristics:
- concise, well structured, illustrated with examples, searchable.

Support methods:

- documentation websites, tutorial videos, onboarding guidance, user communities.

Example: Platforms like AWS provide extensive documentation to simplify complex tasks.

13. Importance of Search and Knowledge Access

Help systems should include **efficient search functionality**.

Capabilities:

- keyword search
- filtering
- suggested answers.

Advantages:

- reduces support load
- improves user autonomy
- speeds up problem resolution.

14. Role of Community Support

User communities can complement official support.

Examples:

- discussion forums
- community Q&A
- peer support groups.

Benefits:

- shared solutions
- knowledge accumulation
- faster troubleshooting.

15. Key UX Design Insight

UX design simplifies complex systems.

Core idea: Transform **complicated information and interactions into intuitive experiences**.

UX design fundamentally involves:

- problem solving, understanding user behavior
- reducing friction in human-technology interaction.

16. Practical Skill Development

Improving efficiency often involves learning **accelerators such as shortcuts**.

Benefits:

- faster task execution, increased productivity, improved workflow efficiency.

Example: keyboard shortcuts in software applications.

17. The Core Takeaways

Important principles to remember:

- UX evaluation frameworks help identify usability problems.
- Nielsen's heuristics are the most widely used usability guidelines.
- Systems must always provide **feedback and transparency**.
- Interfaces should mirror **real-world concepts and user expectations**.
- Users must feel **in control with clear navigation and exit options**.
- Consistency across products and industry standards improves usability.
- Prevent errors whenever possible rather than correcting them afterward.
- Recognition-based interfaces reduce cognitive load.
- Systems should support both beginners and expert users.
- Minimalist design improves focus and reduces decision fatigue.
- Effective error handling requires clear communication and recovery options.
- Accessible help and documentation ensure users can complete complex tasks.
- UX design ultimately focuses on **making complex systems easy to understand and use**.

UX DESIGN - LECTURE 04

1. UX Research – Definition

UX research is the process of **collecting insights about users, their needs, behaviors, and environments** to design better digital products.

Research focuses on understanding:

- who the users are
- what goals they have
- what features they value
- how they use products
- what problems or pain points they face

Goal: Design products that are **useful, usable, and aligned with real user needs**.

2. What UX Research Investigates

UX research gathers insights about several aspects:

Users

- current users
- potential users
- future users

User needs and goals

- tasks users want to complete
- motivations behind product usage
- desired features and content

User pain points

- problems faced with current products
- usability issues
- friction points during tasks

Context of use

- where the product is used
- when it is used
- devices used (mobile, desktop, etc.)

User skill level

- technical knowledge
- experience with similar products
- domain expertise

Personalization

- tailoring design and functionality to user preferences.

3. Competitive and Market Research

UX research also examines the broader market.

Key areas:

- competitor products
- industry trends
- new technologies
- feature comparisons

Purpose:

- identify gaps in the market
- differentiate the product
- maintain competitive advantage.

4. Importance of UX Research

UX research is critical for product success.

Benefits include:

- **Better product usability:** Understanding user behavior allows designers to create **simpler and more effective interfaces**.
- **Increased revenue and market share:** Products that solve real user problems attract more users.

- **Reduced development cost:** Research helps focus development on **high-value features** rather than unnecessary ones.
- **Better decision making:** Research provides **data-driven insights** instead of relying on assumptions or internal debates.
- **Competitive advantage:** Understanding trends and competitor weaknesses helps companies act earlier.

5. Role of UX Designers in Research

UX designers often participate directly in research.

Responsibilities may include:

- conducting research
- supervising researchers
- validating research quality
- implementing research insights into designs.

Direct interaction with users helps designers build **better design intuition**.

6. Types of UX Research

There are two main types of UX research.

6.1 Foundational (Generative) Research

Conducted **early in the product lifecycle**.

Purpose:

- understand users
- identify opportunities
- discover pain points
- shape product strategy.

Common methods:

- interviews
- surveys
- focus groups
- ethnography
- market research.

It supports **long-term product strategy**.

6.2 Evaluative (Experimental) Research

Conducted **after a prototype or product exists**.

Purpose:

- evaluate usability
- optimize existing designs
- test product improvements.

Common methods:

- user testing
- A/B testing
- first-click testing
- usability evaluation.

It supports **short-term improvements**.

7. Importance of User Testing

User testing is one of the most valuable UX research methods.

Key insight:

- Around **85% of usability issues can be identified through testing**.

Despite its importance, only **about 55% of companies conduct user testing**.

User testing helps detect:

- usability problems
- confusing navigation
- feature misunderstandings
- unmet user expectations.

8. User Recruitment for UX Research

Recruiting participants is called **user recruitment**. Participants must represent **real target users**.

They should match characteristics defined in **user personas**.
Possible characteristics include:

- demographics
- product usage habits
- profession
- experience level
- knowledge domain
- personality traits
- device usage
- buying behavior.

9. Sources for Recruiting Participants
There are several recruitment approaches.

- **Personal networks:** colleagues, friends, family.

Fast but may not represent real users.
Recruitment agencies
Professional firms recruit participants matching specific criteria.

- Pros: efficient, structured process.
- Cons: expensive, quality varies.

Online research platforms
Examples include specialized UX research platforms.
Users sign up to participate in research for compensation.

- Advantages: quick recruitment, remote testing.
- Limitation:, difficult for niche target groups.

Advertising
Participants can be recruited through:

- social media ads, classified ads, community posts.
- Ads should include: participation criteria, research description, time requirement, incentives.

Avoid revealing product details.
Social media communities
UX researchers can recruit participants from relevant online groups.

- Benefits: access to niche audiences, community engagement.

Special interest groups
Organizations or communities related to the product domain.

- Examples: professional associations, hobby groups, industry communities.

Public place recruitment
Participants are approached in places where target users are likely to be found.
This is called **guerrilla research**.

- Suitable for: interviews, surveys, quick usability tests.

Research participant panels
A database of people who agreed to participate in future research.

- Sources: website opt-in forms, previous research participants, mailing lists.

Snowball recruitment
Existing participants recommend other suitable participants.

- Advantages: quick, low cost.

10. Participant Screening
Before confirming participation, researchers must verify that the participant matches selection criteria.

- Screening may involve: short interviews, phone calls, questionnaires.

The goal is **ensuring participant suitability**.

The goal is **ensuring participant suitability**.
Product details should usually **not be revealed during screening**.

11. Incentives for Participants
Participants are often compensated.
- Possible incentives: money, gift vouchers, products, refreshments, travel reimbursement.

Compensation encourages participation and respects the participant's time.

12. Definition of User Testing
User testing is a UX research method where:
- users interact with a product or prototype
- researchers observe their behavior
- feedback is collected.

Important principle: The **product is tested**, not the user.
If users struggle, the product design must improve.

13. Types of User Testing

Moderated testing
A researcher guides the session.
Features:
- direct interaction
- probing questions
- observation of user behavior.

Most common approach.

Unmoderated testing
Automated testing conducted through software platforms.
Features:
- users follow automated instructions
- interaction data is recorded automatically.

Used for large-scale testing.

14. Testing Environment
Ideally testing should occur in the **real environment where the product is used**.
Example: Testing a fitness app in a gym.
- However, testing can also occur in: meeting rooms, offices, remote sessions.

Usability labs are not always necessary.

15. User Test Tasks
Participants perform tasks using the product.
- Examples: creating an account, searching for content, configuring settings, completing workflows.

Task design principles:
- **Prioritize tasks:** Start with the most important tasks.
- **Start broad, then specific:** Observe natural navigation before testing specific features.
- **Follow natural workflow:** Tasks should mirror real-life usage sequences.
- **Begin with simple tasks:** Build user confidence before introducing complex tasks.
- **Keep tasks realistic:** Use real-world scenarios.

16. Writing Good Test Tasks
Tasks should be **general and scenario-based**, not instruction-based.

Example:
Poor task: “Go to settings and disable notifications.”
Better task: “Show how you would stop receiving notifications.”
This prevents giving users clues about navigation paths.

17. User Testing Moderation Process

Step 1 — Brief the research team

Explain:

- research objectives
- roles of moderators and observers
- observation rules.

Observers should **not interact with participants** during testing.

Step 2 — Welcome and brief the participant

Actions:

- greet participant
- introduce research team
- explain testing process
- emphasize that the product is being tested.

Participants should:

- think aloud during tasks
- feel free to stop tasks anytime.

Consent forms are usually required.

Step 3 — Moderation during testing

Moderator responsibilities:

- assign tasks
- encourage thinking aloud
- ask probing questions
- observe behavior
- record observations.

Important observations:

- hesitation
- confusion
- frustration
- unexpected navigation.

If a participant becomes stuck, the moderator may assist to continue testing other tasks.

Step 4 — Post-test questions

After tasks are completed, ask participants:

- what they liked
- what they disliked
- suggestions for improvement.

Common metric: Net Promoter Score (NPS)
Example question: “How likely are you to recommend this product to others?”

Step 5 — Debrief the participant

Actions:

- thank the participant
- provide incentive
- ask if they are willing to participate in future studies.

Participants may be added to a research panel.

Step 6 — Debrief the research team

Immediately after the session:

- summarize observations
- capture insights

- record usability problems.

Delaying this step may cause important insights to be forgotten.

18. Key Takeaways

Important concepts to remember:

- UX research gathers insights about users, needs, and behaviors.
- Research aligns product design with **real user needs**.
- Two main research types exist: **generative research and evaluative research**.
- User testing is the most common UX research technique.
- Recruiting the right participants is critical to valid research.
- Participants must represent real target users.
- UX testing focuses on evaluating the **product**, not judging the user.
- Effective test design includes realistic tasks and careful observation.
- Moderated sessions involve briefing, testing, feedback, and debriefing.
- Immediate documentation of findings ensures valuable insights are preserved.

UX DESIGN - LECTURE 05

1. User Flow – Definition

A **user flow** is the path a user follows when interacting with a digital product such as an app, website, or system.

It describes:

- where the user **starts**
- the **steps or screens** they move through
- how they **reach their final goal**

User flow essentially defines the **navigation logic of the product**.

2. Purpose of User Flow in UX Design

User flows help designers understand and structure how users interact with a system.

Key purposes:

- plan navigation before building the interface
- visualize the structure of the application
- map user interactions step-by-step
- identify potential usability barriers
- support logical interface design.

3. Elements Considered in User Flow Design

When designing a user flow, UX designers consider the following factors.

User objectives: What users want to achieve using the application.

Engagement flow: How users move through the interface while interacting with features.

Information requirements: What information users need during navigation.

- Examples: alerts, hints, visual cues, instructional text.

Barriers: Anything preventing users from reaching their final goal.

- Examples: confusing navigation, missing information, unnecessary steps.

4. Advantages of User Flow Diagrams

User flows provide several benefits in the design process.

- **Visual communication:** They clearly illustrate how the application works.
- **Logical structure planning:** Designers can define interaction logic before development.
- **Navigation mapping:** Helps identify how users move between screens.
- **Early feedback:** Stakeholders can review flows before prototyping begins.
- **Time savings:** Design issues can be fixed before building the prototype.

5. Steps to Create a User Flow

The UX design process for creating a user flow generally follows these steps.

Step 1 — Profile the users

Identify who will use the application.

This includes:

- user characteristics
- behaviors
- needs.

Step 2 — Define user goals

Understand what users want to achieve.

Example goals:

- finding information
- completing a purchase
- learning about a product.

Step 3 — Identify information needs
Determine what information users need to complete tasks.

- Examples: instructions, notifications, interface cues.

Step 4 — Create visual user flow diagrams
Represent the navigation steps visually using flow diagrams.

- These diagrams map: screens, interactions, transitions.

Step 5 — Develop the prototype
Convert the flow diagram into a **prototype**.

- Prototypes can be either low fidelity or high fidelity.

The prototype transforms the **conceptual flow into a working interface simulation**.

6. User Flow Diagram Architecture
A **user flow diagram** visually represents the structure of the application.
Common diagram components include:

- **Start node:** Entry point of the application.
- **Screens / processes:** Represent individual screens or actions in the system.
- **Arrows:** Show navigation between screens.
- **Decision nodes:** Used when conditional logic exists.

Example: Yes / No decisions.

7. Linear vs Conditional User Flows
User flows can follow different structures.

Linear flows
Users move through screens in a fixed sequence.
Example: Start → Screen A → Screen B → Screen C → End
No branching decisions.

Conditional flows
Include decision points where users choose different paths.
Example: Start → Decision, Yes → Path A, No → Path B
Decision nodes are usually represented using **diamond shapes**.

8. Example User Flow – Plant Information App
The lesson example uses a simple plant information app called **Planto**.
Purpose of the app: Provide information about different plant species.

- Features include: plant categories, plant details, augmented reality viewing.

9. Example User Flow Structure
The application flow is linear and consists of the following steps.

1. Start screen (splash screen)
2. Home screen
3. Categories screen
4. Plant type screen
5. Plant details screen
6. AR mode screen
7. End

This represents the **complete interaction journey**.

10. Screen Functions in the Example Flow

- **Start screen:** Entry point when the app launches.
- **Home screen:** Contains a **Get Started / Explore** button.
- **Categories screen:** Allows users to select plant categories.

    - Examples: succulents, vegetables, house plants.
- **Plant type screen:** Displays plant species within a selected category.
- **Plant details screen:** Shows information about the selected plant.
    - Examples: plant type, sunlight requirement, moisture tolerance.
- **AR mode screen:** Allows users to visualize the plant in their real environment using augmented reality.
- **End state:** Final stage of the interaction flow.

11. Tools for Creating User Flow Diagrams

User flow diagrams can be created using diagramming tools.

Example tool used in the lesson: **draw.io**

Capabilities:

- flowchart creation, UX diagrams
- system architecture diagrams.

Storage options:

- local device
- cloud services (Google Drive, OneDrive).

12. Basic Flowchart Elements Used

Common flowchart symbols used in UX flows include:

- **Start / End symbol:** Represents beginning or completion.
- **Rectangle:** Represents screens or processes.
- **Arrow connectors:** Represent navigation paths.
- **Diamond:** Represents decisions or conditional logic.

13. Purpose of Creating User Flows Before Prototyping

User flows are created before building prototypes to:

- define interaction logic
- ensure logical navigation

- define interaction logic
- ensure logical navigation

This step ensures the prototype follows a **clear user journey**.

14. Relationship Between User Flow and Prototyping

User flow diagrams act as the **foundation for prototypes**.

Process relationship: User Flow → Wireframes → Prototype → Testing.

User flow defines **structure**, while prototypes define **interface behavior**.

15. Key Takeaways

Important points to remember:

- A user flow maps the path users take through an application.
- It defines the **start point, interaction steps, and end goal**.
- User flows help plan navigation before building prototypes.
- They identify barriers that might prevent users from achieving goals.
- User flow diagrams visually represent application structure.
- Linear flows follow a fixed sequence of steps.
- Conditional flows include decision points.
- Creating user flows early reduces redesign effort later.
- Flow diagrams act as the **logical blueprint for prototypes and interface design**.

UX DESIGN - LECTURE 06

1. Prototyping in UX Design
Prototyping is the process of creating a **working model of a digital product** to demonstrate how the final application will function.
Purpose:
- visualize the product before development
- test ideas quickly
- present the concept to stakeholders and users
- identify design problems early.

A prototype is **not the final product**, but a simulation of how the product will behave.

2. Role of Prototypes in the UX Process
Prototypes are created after planning stages such as:
- user research, user flows, wireframes.

They help translate design ideas into **interactive experiences** that stakeholders and users can explore.
- Benefits: reduces development risk, allows quick design changes, improves communication with stakeholders.

3. Prototyping Tools
Various tools are used to create UX prototypes.
Example used in this lesson: **Figma**
Characteristics of Figma:
- cloud-based design tool
- collaborative environment
- supports UI design and prototyping
- allows sharing designs with teams.

Other capabilities:
- real-time collaboration
- design libraries
- plugins and widgets.

4. Figma Environment Overview
The Figma interface consists of several main areas.
- **Navigation panel:** Used for organizing design files.
  - Contains: recents, drafts, teams, community resources.
- **Design canvas:** The main workspace where screens and UI elements are created.
  - Supports: zooming, panning, arranging design elements.
- **Toolbar:** Provides tools for creating elements such as:
  - frames, shapes, text, vectors, images.
- **Layers panel:** Displays all elements placed on the canvas.
  - Used for: selecting objects, organizing components, managing hierarchy.
- **Properties panel:** Allows editing properties of selected objects.
  - Examples: color, size, positioning, border radius.

5. Creating a New Design File

To begin prototyping in Figma:

1. Open **figma.com**
2. Select **New Design File**
3. A new canvas workspace is created.

This workspace will contain the screens for the prototype.

6. Frames in Figma

A **frame** represents a screen in the application.

Frames are used to simulate devices such as:

- mobile phones
- tablets
- desktops.

Examples of available presets:

- Android devices
- iPhone devices
- desktop screens.

Each screen in the app is designed inside its own frame.

7. Example Prototype Project

The lesson demonstrates building a prototype for a simple plant information app called **Planto**.

The app provides:

- plant categories
- plant species information
- plant care details
- augmented reality visualization.

The first stage of the prototype focuses on creating **two screens**.

8. Screen 1 — Splash Screen

The splash screen is the **entry point** of the application.

Elements created:

- **Background layer:** A rectangle covering the entire frame used as the screen background.
- **Color palette:** Colors are selected using a color palette tool. Color palettes are defined using **hex codes**.
  - Example: Primary color applied to the background.
- **Icon:** An icon representing the app is added.
  - Source: Google Material Icons.
  - Preferred file format: **SVG**
  - Advantages (SVG): scalable without losing quality, editable colors, vector-based graphics.
- **Application logo text**
  - Text added using the text tool.
  - Example: App name “Planto”.
  - Font used: Roboto Slab.
  - Text alignment ensures the logo is **perfectly centered on the screen**.

9. Color Palette Selection

Color palettes help maintain **consistent visual identity**.

Example tool used: Material palette generator.

Palette components include:

- primary color
- light primary color
- dark primary color
- accent color.

Hex codes allow designers to apply colors consistently across the interface.

10. Screen 2 — Home Screen
The second screen represents the **main entry interface of the app**.
This screen is created by duplicating the splash screen frame and modifying it.
Key elements include:

- **Background transition design**
  - A curved design created using: ellipse shape, rectangle shape, boolean subtraction.
  - This creates a visual transition between screens.
- **Logo and icon repositioning:** The app logo and icon move to the top section of the screen.
- **Call-to-action button:** A **Get Started** button is added.
  - Button components: rectangle shape, text label.
  - Button design improvements: rounded corners, centered text, consistent color styling.

11. Button Design Principles
Buttons are designed with several UX considerations.

- Elements include: color contrast, rounded corners, clear label text.

Rounded corners improve modern UI appearance.
Example setting: Border radius applied to soften edges.

12. Grouping Elements in Figma
Multiple elements can be grouped together.
Purpose:

- move related components together
- simplify layer management.

Example grouping:

- button shape
- button text.

Grouped elements behave as a single component.

13. Adding Interactions in Prototypes
Interactions simulate how users interact with the interface.

- Example interactions include: tapping, scrolling, dragging.

In this lesson, the prototype interaction is:
Tap → Navigate to another screen

14. Navigation Between Screens
To create navigation:

1. Select the source screen
2. Open the **Prototype tab**
3. Add interaction rule
4. Define action:
   - On tap
   - Navigate to destination screen.

This creates a visual connection between screens.

15. Smart Animate
Figma provides animation effects for transitions.
Example used: **Smart Animate**
Purpose:

- smoothly animate changes between screens
- create natural movement between elements.

Animation properties include:

- easing style
- duration.

Example duration: 300 milliseconds.

16. Testing the Prototype
Figma allows designers to run the prototype.

Process:

- click **Present / Play**
- the interface runs as an interactive simulation.

Users can click through screens and experience the interaction flow.

17. Advantages of Prototyping with Figma

Figma provides several benefits for UX designers.

Advantages include:

- cloud collaboration
- real-time editing
- easy sharing with stakeholders
- quick iteration of designs.

Prototypes help teams validate ideas **before development begins**.

18. Relationship Between User Flow and Prototyping

User flows define the **logical path of interaction**.

Prototypes convert that flow into **interactive screens**.

Process sequence:

User Flow → UI Design → Prototype → User Testing.

The prototype visually implements the navigation structure defined earlier.

19. Key Takeaways

Important concepts to remember:

- Prototyping creates interactive representations of digital products.
- Figma is a cloud-based tool used for UI design and prototyping.
- Frames represent individual screens of an application.
- SVG icons are preferred because they are scalable and editable.
- Color palettes help maintain consistent visual design.
- UI elements include backgrounds, icons, text, and buttons.
- Interactions define how users move between screens.
- Smart animation improves transitions between interface states.
- Prototypes allow testing and feedback before development begins.

UX DESIGN - LECTURE 07

Building Additional Screens & Implementing User Flows in Figma

1. Lesson Focus

This lesson continues the development of the **Planto mobile prototype** inside Figma.

Main focus:

- working with **elements**
- manipulating **element properties**
- implementing **user flows and navigation**
- building additional **application screens**

Two major screens are built in this lesson:

1. **Category Screen**
2. **Plant Type Screen**

2. Application Flow Context

From the earlier user flow diagram, the application contains these screens:

1. Splash Screen
2. Home Screen
3. Category Screen
4. Plant Type Screen
5. Plant Details Screen
6. AR Mode Screen

This lesson focuses on creating:

Category Screen → Plant Type Screen

3. Creating the Category Screen

Step 1 — Create a Frame

Create a new mobile frame > Frame preset: Android Small > Rename frame: **category screen**

This screen will display the different plant categories.

4. Adding Navigation Elements

Back Arrow

A back arrow icon is imported from **Google Material Icons**.

Steps:

1. Search for **arrow back**
2. Download as **SVG**

Import into Figma using: File → Place Image

SVG format is used because:

- it scales without losing quality
- colors can be modified

Page Title

A heading is added next to the back button.

Text: Explore

Purpose: This title communicates that users are exploring plant categories.

- Font: Source: Sans Pro, Size: 30px

5. Creating Category Cards

The category screen displays **three category cards**:

- Succulents, Vegetables, House Plants

Each category is represented as a **visual card**.

6. Designing a Category Card

Card Container: Create a rectangle.

Dimensions: 355 × 170 px

This rectangle acts as the **card holder**.

Adding Image

Insert a plant image representing the category.

Example: succulents image

- Steps: Place image > Scale using Shift > Use **crop** to fit inside the card

Dark Overlay Mask
A semi-transparent mask is placed on top of the image.
Color: #282727
Opacity: 40%

Purpose:
- improves text visibility
- creates a modern UI style

Animated Vertical Line
A vertical colored line is added.
Line width: 5px
Color: dark green
Purpose: Used later for **hover animation feedback**.

Category Title
Add the category name.
Example: Succulents
- Text style: SemiBold, Size: 24px

This visually distinguishes it from the main page heading.

7. Organizing UI Elements
Each element inside the card is renamed:
Examples:
- succulents line
- succulents mask
- succulents image
- succulents holder

Then grouped as:
- succulents card

Proper naming helps maintain **clean project structure**.

8. Reusing UI Components
Instead of designing new cards from scratch:
Use duplication > Shortcut: Ctrl + D
The first card is duplicated twice to create:
- Vegetables card
- House plants card

Images and text are then replaced accordingly.

9. Adding Depth with Shadows
To make the UI feel more realistic:
Apply **Drop Shadow**.
Opacity: 35%
Benefits:
- adds visual depth
- separates cards from background

10. Linking Screens (Navigation Flow)
Navigation is implemented using Figma's **Prototype tab**.
Example:
- Interaction: On Tap → Navigate To → Category Screen
- Animation: Slide In
- This connects: Home Screen → Category Screen

11. Creating Hover Animation
To provide interaction feedback:
A second version of the category screen is created.
Duplicate screen: category screen → category screen anim
Changes in animated version:
- vertical line expands
- image zooms slightly

Animation Changes
Line Expansion: Line width increased from: 5px → 12px

Image Zoom: Image is scaled slightly larger and recropped.
Purpose: Create **visual feedback** when user interacts.

12. Triggering Animation
Interaction settings: While Hovering → Open Overlay
Effect: Dissolve
This produces a smooth hover animation effect.

13. Creating the Plant Type Screen
A new frame is created.
Frame name: plant type
This screen displays **individual plant species** within a category.
Example category: Succulents

14. Reusing Existing UI Elements
Instead of rebuilding elements:
- Copy: back arrow & page title

Rename title: Succulents
This maintains **design consistency**.

15. Creating Plant Cards
Plant cards show individual species.
- Example plants: Agave, Black Rose, Red Aloe

Card Layout
Card dimensions: 143 × 170 px
Features:
- rounded corners
- drop shadow
- plant image
- plant name

Corner radius: 15px

16. Adding Plant Images
Example: Agave plant
Image is placed and scaled to fit.
Top corners are rounded using **independent corner radius**.

17. Adding Plant Name
Example text: Agave
Additional description:
- Shark Skin Succulent
- Color: dark green (#19571C)

18. Adding Search Bar
A search bar is placed above the plant cards.
Components:
- rectangle container
- rounded edges
- search icon

Radius: 25px
Search icon is imported from Google Icons and recolored green.

19. Grouping Plant Cards
Each plant card includes:
- image
- card holder
- plant name text

Grouped as:
agave card, black rose card, red aloe card
Cards are duplicated using **Ctrl + D** to speed up design.

20. Adding Scan Button
A button is added at the bottom.
Label: Scan
Purpose: Demonstrates a **future feature idea**.
Possible feature: Scanning a plant using the camera to identify it.
Even if functionality isn't implemented, prototypes can show **concept features**.

21. Connecting Category → Plant Type
Interaction added to: succulents card
Prototype interaction: On Tap → Navigate To → Plant Type Screen
Animation: Smart Animate

22. Resulting Navigation Flow
Current prototype flow: Splash Screen > Home Screen > Category Screen > Plant Type Screen
Users can now:

- open categories
- view plant types
- interact with animated cards

23. Key UX Concepts Demonstrated
Important UX principles shown in this lesson:

- **Reusable Components:** Reuse UI modules instead of redesigning them.
- **Visual Feedback:** Hover animations indicate interaction.
- **Hierarchy:** Different text sizes distinguish headings and subheadings.
- **Clean UI Structure:** Proper naming and grouping improve workflow.
- **Rapid Prototyping:** Design features even if functionality is not implemented.

24. Next Step in the Application
The next lesson will complete the prototype by adding:

- **Plant detail screen**
- **Augmented Reality (AR) viewing feature**

The AR feature will allow users to visualize plants inside their environment.

UX DESIGN - LECTURE 08

Completing the UX Prototype (Planto App)
This lesson completes the **first full prototype** built in Figma.
The final tasks include:

- building **plant detail screens**
- implementing the **remaining user flows**
- adding an **AR (Augmented Reality) feature**
- implementing **back navigation**

At the end of this lesson, the entire application flow becomes functional.

1. Final Application Structure
By the end of the prototype, the application contains these screens:

1. Splash Screen
2. Home Screen
3. Category Screen
4. Plant Type Screen
5. Plant Detail Screens
6. AR Mode Screen

- Navigation flow: Splash → Home → Categories → Plant Type → Plant Detail → AR Mode

Back navigation allows users to move backwards through the flow.

2. Creating the Plant Detail Screen
A **plant detail screen** displays specific information about an individual plant species.
Example plant: Agave (Shark Skin Succulent)
Step 1 — Create a Frame
Create a new frame using > Frame → Android Small > Rename frame: agave plant

3. Plant Image Section
Add a rectangle container to hold the plant image.
Example dimensions: 325 × 254 px
Then insert the plant image using: File → Place Image
Example image used: Agave
This large image acts as the **hero image** of the detail screen.

4. Plant Title
Add the plant name below the image.
Example: Agave Shark Skin
Formatting:

- Font: Source Sans Pro
- Font size: 20
- Style: Bold
- Color: Dark green theme

Purpose: The title clearly identifies the plant species.

5. Plant Information Card
Below the plant title, a **white information card** is created to display plant attributes.
Rectangle dimensions: 328 × 189 px
Design styling: rounded corners, drop shadow, white background
Corner radius: 15 px
Drop shadow opacity: 35%
This creates a clean **content container**.

6. Plant Attribute Icons
Icons are imported from **Google Material Icons**.
Three icons are used:

1. Description icon
2. Water drop icon
3. Sun / light icon

Icons represent key plant attributes.
Icons are resized to around:~31–35 px

Example mapping:

Icon	Document	Water drop	Sun
Attribute	Type	Moisture	Light

7. Plant Attribute Labels
Text labels are added next to each icon.
Example labels: Type. Moisture. Light
Font settings:

- Font: Source Sans Pro
- Size: 16
- Color: Green theme

8. Plant Attribute Values
Descriptions are added below each label.
Example values for **Agave**:
Type: Succulent
Moisture: Drought tolerant
Light: Full sun
Font settings: Font size: 12, Style: Regular
This layout creates a simple **information list**.

9. Adding the AR Button
A button is added at the bottom of the screen.
Text: View in Room
Purpose:

- This button opens **Augmented Reality Mode** to visualize the plant.
- The button is reused from earlier designs to maintain UI consistency.

10. Duplicating Plant Detail Screens
Instead of redesigning screens, the Agave screen is duplicated > Shortcut: Ctrl + D
Two additional screens are created: **Black Rose Plant, Red Aloe Plant**

Changes required:

- update plant name
- replace plant image
- adjust moisture description if needed

Example: Black Rose
Moisture: Dry to medium
Light: Full sun

11. Connecting Plant Type → Plant Detail
Prototype interactions are added.
Example: On Tap → Navigate To → Agave Plant Screen

- Similar flows are created for: Black Rose, Red Aloe

This connects the plant list to the detailed view.

12. Creating AR Mode Screen
A new frame is created. Name: AR Mode
Basic elements include: back arrow, title, animated visual

13. Adding AR Simulation
Figma **does not support video formats** like: MP4, MKV, AVI
Instead, motion is simulated using: GIF files
GIFs are treated as images but display animated frames.
Steps: File → Place Image → Insert GIF
The GIF represents the plant being viewed in **Augmented Reality mode**.

14. Adjusting AR Media

- If needed: rotate the image, scale it, send it behind UI layers

Rotation shortcut: Ctrl + Shift + Drag
The back arrow remains on top for navigation.

15. Connecting AR Button
Interaction for the button:
On Tap → Navigate To → AR Mode Screen
Animation: Slide In
This makes the AR screen appear naturally during navigation.

16. Implementing Back Navigation
Back arrows are added across screens.
Interaction setting: On Tap → Back
The **Back action** returns the user to the previous screen in the navigation stack.

- This is applied to: plant detail screens, AR screen

17. Final Prototype Flow
Complete navigation flow: Splash Screen > Home Screen > Category Screen > Plant Type Screen > Plant Detail Screen > AR Mode
Back navigation allows reverse movement through each step.

18. Demonstrating the Prototype
When the prototype is run:

1. Splash screen appears
2. Tap → Home screen
3. Tap "Get Started"
4. Choose plant category
5. Select plant type
6. Open plant details
7. Tap "View in Room"
8. AR mode simulation appears

19. Prototyping Tip for UX Designers

Even if functionality is not implemented, prototypes should **demonstrate potential features**.

Examples:

- AR product visualization
- plant scanning
- AI recommendations

This helps communicate product vision to:

- stakeholders
- investors
- developers
- users

20. Important UX Lessons From This Prototype

1. **Consistent design system:** Reusing components maintains visual consistency.
2. **Structured naming:** Proper naming improves collaboration with developers.
3. **Interaction feedback:** Animations and transitions improve usability.
4. **Modular design:** Duplicating components saves time.
5. **Feature demonstration:** Prototypes can showcase features that don't yet exist.

21. Final Result

The **Planto prototype** now demonstrates:

- complete navigation
- category browsing
- plant exploration
- detailed information
- AR visualization concept

This concludes the **first full UX prototype project**.

The next prototype project will focus on a **more advanced e-commerce application**.

RESOURCES/REFERENCES

Basics (UI/UX Fundamentals)

- https://lawsofux.com
- https://uxplanet.org
- https://designbetter.co
- https://www.uxtools.co
- https://www.nngroup.com
- https://uxdesign.cc
- https://learnui.design
- https://refactoringui.com
- https://www.uxbeginner.com
- https://www.interaction-design.org
- https://www.uxtools.co

UX Courses & Structured Learning

- https://www.coursera.org/professional-certificates/google-ux-design
- https://careerfoundry.com/en/courses/become-a-ux-designer
- https://www.udacity.com/course/ux-designer-nanodegree--nd578
- https://www.springboard.com/courses/ui-ux-design-career-track
- https://www.codecademy.com/learn/ux-design

UX Case Studies

- https://growth.design/case-studies
- https://www.behance.net/galleries/ui-ux/case-study
- https://uxdesign.cc/tagged/case-study

UI Inspiration

- https://www.awwwards.com
- https://godly.website
- https://land-book.com
- https://calltoinspiration.com
- https://www.siteinspire.com
- https://www.lapa.ninja
- https://onepagelove.com
- https://minimal.gallery

Real Product UI (Best References)

- https://mobbin.com
- https://screenlane.com
- https://uxarchive.com
- https://pageflows.com
- https://www.pttrns.com
- https://uisources.com

SaaS / Startup UI

- https://saasframe.io
- https://saaspo.com
- https://landingfolio.com
- https://cosmos.so

UX Patterns & Components

- https://www.uxlibrary.org
- https://ui-patterns.com
- https://www.checklist.design
- https://goodui.org

Design Systems

- https://designsystemsrepo.com
- https://www.designsystems.com
- https://component.gallery
- https://storybook.js.org/showcase

Micro Interactions / Animations

- https://www.hoverstat.es
- https://lottiefiles.com
- https://www.animista.net
- https://ui.aceternity.com

Design Resources

- https://www.figma.com/community
- https://www.figmacrush.com
- https://www.figma.com/community/libraries
- https://flowbase.co
- https://www.uistore.design

This is an awesome list on GitHub for UI/UX resources. Its last update was in 2024, so as of 2026, some links may be slightly outdated.

- https://github.com/kevindeasis/awesome-ui

All related information and resources can be found at the following link:

Github > Runarok > ux-game-intern